

Energijski zank

$$dU = dQ + dW \Rightarrow d\bar{E} = dQ - p dV \quad / \cdot \beta$$

$$\bar{E} = \frac{d(\beta F)}{d\beta}$$

$$d(\beta F) = \bar{E} d\beta - p d(\beta V) = \bar{E} d\beta - p \beta dV$$

$$p = - \left(\frac{\partial F}{\partial V} \right)_\beta = - \left(\frac{\partial (\beta F)}{\partial (\beta V)} \right)_\beta$$

$$\left. \begin{array}{l} \beta d\bar{E} = \beta dQ - p \beta dV \\ d(\beta F) = \bar{E} d\beta - p \beta dV \end{array} \right\} \quad \begin{array}{l} \beta d\bar{E} - d(\beta F) = \beta dQ - \bar{E} d\beta \\ d(\beta \bar{E}) - d(\beta F) = \beta dQ \end{array}$$

$$\beta dQ = d(\beta \bar{E} - \beta F) \quad / \cdot k_B$$

$$dS = \frac{dQ}{T} = k_B (\beta \bar{E} - \beta F)$$

$$S = k_B \beta (\bar{E} - F) = \frac{1}{T} (\bar{E} - F)$$

TD: prosta energija $F = U - TS \rightarrow S = \frac{\bar{E} - U + F}{T}$

$$S = k_B \beta (\bar{E} - F) = k_B \overbrace{\beta (\bar{E} - F)}^{-\ln g_c} = -k_B \overline{\ln g_c} = \boxed{-k_B \int p \ln g_c d\Gamma}$$

GIBBSOVA FORMULA
ZA ENTROPIJO

Mikrokanonična porazdelitev:

$$p(\epsilon) = \begin{cases} \frac{1}{\Delta\epsilon} & \epsilon_0 - \frac{\Delta\epsilon}{2} < \epsilon < \epsilon_0 + \frac{\Delta\epsilon}{2} \\ 0 & \text{sicer} \end{cases}$$

$$\rightarrow S = -k_B \int \frac{1}{\Delta\epsilon} \ln \frac{1}{\Delta\epsilon} d\Gamma = -k_B \frac{1}{\Delta\epsilon} \ln \frac{1}{\Delta\epsilon} \Delta\epsilon$$

$$\boxed{S = k_B \ln(c \Delta\epsilon)} \quad \text{Boltzmannova formula za entropijo}$$

Idealni atomni plin

$$e^{-\beta F} = C V^N \left(\frac{2\pi m}{\beta} \right)^{3N/2}$$

$$F = -k_B T \left(\ln C + N \ln V + \frac{3N}{2} \ln \frac{2\pi m}{\beta} \right)$$

$$\bar{E} = \frac{3}{2} N k_B T$$

$$S = \frac{1}{T} \left[\frac{3}{2} N k_B T + k_B T \ln C + k_B N T \ln V + \frac{3}{2} N k_B T \left(\ln 2\pi m \downarrow \frac{k_B T}{m} \right) \right] =$$

$$= N k_B \ln V + \frac{3}{2} N k_B \ln T + \text{konst.} = \overset{\text{masa plina}}{\frac{m^* N_A}{M}} k_B \ln V + \frac{3}{2} \overset{\text{masa atoma}}{\frac{m^* N_A}{M}} k_B \ln T + \text{konst.} =$$

$$= \frac{m^* R}{M} \ln V + \frac{3}{2} \frac{m^* R}{M} \ln T + \text{konst.}$$

iz TD: $\Delta S = m^* C_V \ln \frac{T}{T'} + \frac{m^* R}{M} \ln \frac{V}{V'}$

Boltzmann $\rightarrow$ Gibbs

N klas. delcev

k stanj

• konfiguracija: $\{N_1, N_2, \dots, N_k\}$

$$S(\{N_1, \dots, N_k\}) = k_B \ln \Omega$$

$\rightarrow$ št. stanj: multinomski simbol:

$$S = k_B \ln \frac{N!}{N_1! N_2! \dots N_k!} = k_B [\ln N! - \ln N_1! - \dots - \ln N_k!]$$

$$= k_B \left[\overset{N_1 + \dots + N_k}{N} \ln N - \overset{N_1}{N_1} \ln N_1 - \dots - \overset{N_k}{N_k} \ln N_k \right]$$

$$= k_B \left(N_1 \ln \frac{N}{N_1} + N_2 \ln \frac{N}{N_2} + \dots + N_k \ln \frac{N}{N_k} \right) =$$

$$= -N k_B \left(\frac{N_1}{N} \ln \frac{N_1}{N} + \dots + \frac{N_k}{N} \ln \frac{N_k}{N} \right) =$$

$$= -N k_B \sum_{j=1}^k \frac{N_j}{N} \ln \frac{N_j}{N} = -N k_B \sum p_j \ln p_j \Rightarrow \underline{S = -N k_B \int p \ln p \, d\Gamma}$$

verjetnost, da delec v j-tem predelčku

$$\frac{N!}{N_1! N_2! \dots N_k!}$$

$$\sum_{j=1}^k N_j = N$$

STIRLING:

$$\ln N! \approx N \ln N - N$$

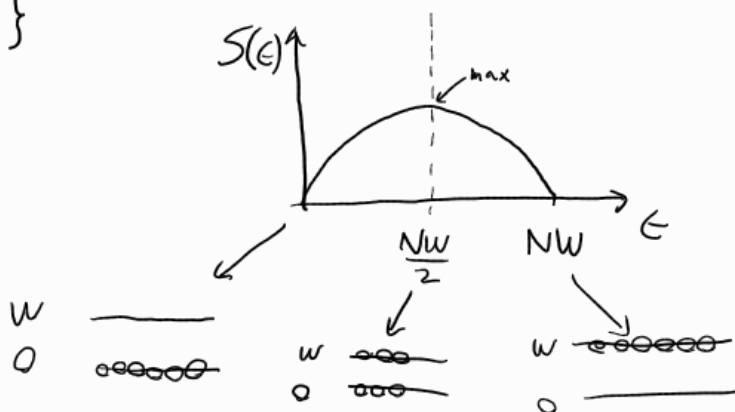
Dvonivojski sistem

W ————— N'
 0 ————— $N-N'$
 N delcev

• energija: $E = N'W$

• konfiguracija: $\{N-N', N'\}$

$$\begin{aligned}
 S(N') &= S(E = N'W) = \binom{N}{N'} = \binom{N}{N-N'} \\
 &= k_B \ln \Omega = k_B \ln \frac{N!}{(N-N')! N'!} \\
 &= k_B \ln \frac{N!}{(N - \frac{E}{W})! (\frac{E}{W})!}
 \end{aligned}$$



• pri E_{min} in E_{max} imamo le en način, da dobimo to stanje (vse dano v zg./sp. stanje) $\Rightarrow S = k_B \ln 1 = 0$

• največ načinov dobimo pri $\frac{NW}{2}$ (Pascalov 3kotnik)

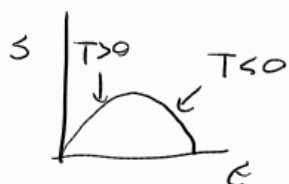
$$\begin{aligned}
 S &= k_B \left[N \ln N - N \left(N - \frac{E}{W} \right) \ln \left(N - \frac{E}{W} \right) + N \frac{E}{W} - \frac{E}{W} \ln \frac{E}{W} + \frac{E}{W} \right] = \\
 &= N k_B \left[\left(1 - \frac{E}{NW} \right) \ln N + \frac{E}{NW} \ln N - \left(1 - \frac{E}{NW} \right) \ln \left(N - \frac{E}{W} \right) - \frac{E}{NW} \ln \frac{E}{W} \right] = \\
 &= -N k_B \left[\left(1 - \frac{E}{NW} \right) \ln \left(1 - \frac{E}{NW} \right) + \frac{E}{NW} \ln \frac{E}{NW} \right]
 \end{aligned}$$

$$dE = T ds + d\tilde{W}$$

$$T = \frac{dE}{ds} = \left(\frac{ds}{dE} \right)^{-1}$$

$$\frac{ds}{dE} = -N k_B \left[-\frac{1}{NW} \ln \left(1 - \frac{E}{NW} \right) - \frac{1}{NW} \frac{\left(1 - \frac{E}{NW} \right)}{\left(1 - \frac{E}{NW} \right)} + \frac{1}{NW} \ln \frac{E}{NW} + \frac{1}{NW} \right] =$$

$$= \frac{k_B}{W} \left[\ln \left(1 - \frac{E}{NW} \right) - \ln \frac{E}{NW} \right] = \frac{k_B}{W} \ln \frac{1 - \frac{E}{NW}}{\frac{E}{NW}} = \frac{k_B}{W} \ln \left(\frac{NW}{E} - 1 \right)$$



$$\frac{\rho(E + \Delta E)}{\rho(E)} = \frac{\exp(-\beta(E + \Delta E) - F)}{\exp(-\beta(E - F))} = \exp(-\beta \Delta E)$$

